Jolla tab coming
Jolla has passed its crowdfunding goal of $380k in Indiegogo of building open source iPad alternative Sailfish OS tablet **http://dgit.in/JollaTab**

Audi's Hydrogen cars
Audi will soon be launching its Hydrogen version cars with the A7, that could go 300 miles on a tank **http://dgit.in/AudiHCar**

laws. Some critics have even questioned the parameters of the experiment, claiming that the observed results may have simply been an error in experiment design or some overlooked parameter. After all NASA has done it before, when it claimed to have successfully tested an anti-gravity device which just turned out to be interference affecting the measurement instruments.

In light of NASA's own controversial history, when it comes to such experiments we can assume that the government agency would have to be very certain about the results before releasing them to the public. On the other hand, it would've been a lot easier to dismiss NASA's claims as premature and not clearly understood if it was the first organisation to attempt such a device. Interestingly, it's the third group to have made such claims, and being the most reputed has garnered the most attention. The first two experimental claims came from respected British scientist Roger J. Shawyer and Juan Yang, Professor of Propulsion Theory and Engineering of Aeronautics and Astronautics in China.

The many who came before
The earliest and most trusted source of the idea for a reactionless propulsion engine came from Roger J. Shawyer in 2006. It was the idea for the EmDrive, also known as the RF (radio frequency) resonant cavity thruster. The design of the original EmDrive was similar to the one tested by NASA in July 2014, and it made use of the tapered resonant cavity powered by a magnetron generating microwaves. He claimed that in experiments, there was a successful conversion of 300 watts of power to 85 millinewtons of thrust.

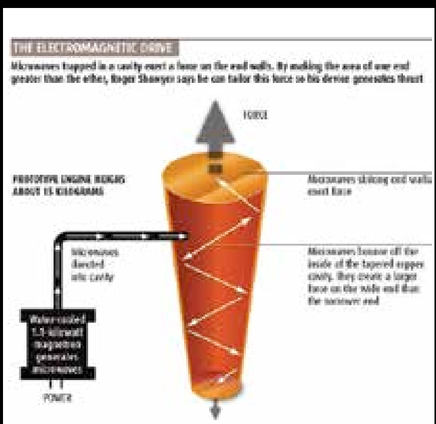
The news made headlines with many hailing it as a breakthrough in Propulsion Engineering. However, since it was a part of a private venture, it couldn't be academically investigated or verified by outside agencies. Shawyer has continued his claims that the device doesn't violate the laws of conservation of momentum and energy. Instead, he attributes the thrust to be the result of imbalance in 'radiation pressure' between the two faces of the device. Till date, there has been no verified explanation or understanding of this rationale by scientists.

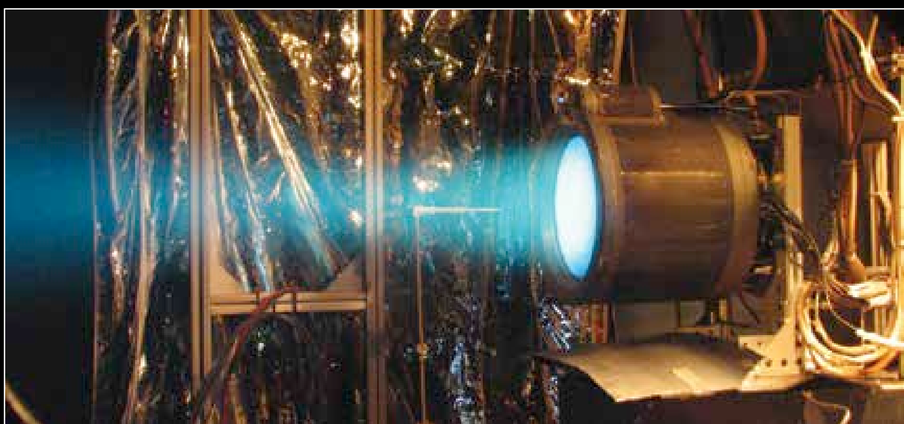
Another similar experiment was conducted by Juan Yang at the Northwestern Polytechnical University in Xi'an, China in 2010. Yang used the same design as the EmDrive and claimed that her experiment was a massive success. Her claims stated that in terms of quantum theory, it was possible to achieve 456 millinewtons of thrust from 1 kilowatt of power and that she had achieved 720 millinewtons from 2.5 kilowatts of power in experiments. However, due to the relatively closed nature of Chinese academia there has been no successful validation beyond the publicly released research papers, which in themselves raise more questions than they answer.

So far, all the publicly available experiments attribute their results as a variation of quantum theory of relativistic propulsion. But, we know that there's far too wide a knowledge gap in the field of quantum mechanics for there to be any meaningful consensus among experts. We should remember that science relies on replicable, observable, explainable and experimental results, and not just a majority consensus amongst scientists. In light of these realities, we should exercise caution in accepting the experiments results from NASA, Shawyer and Yang and not view them as set in stone. But, we really do wish they prove to be true.

The Tyranny of the Rocket Equation
In its research paper, NASA declared that the expected ratio of thrust-to-power for its Cannae drive powered engines in flight applications would be about 0.4 newtons per kilowatt of electric power. To put this in context, the state-of-the-art thrusters used in orbit today require only .06 newtons per kilowatt – which is seven times lesser than what NASA claims possible. This means that if the technology for such reactionless engines can be harnessed, then we can significantly cut the costs and difficulties of space travel.

The dynamics of space travel are built on simple Newtonian physics. The third law of which states that in a closed system, every action has an equal and opposite reaction. We study these equations in school and can see how they apply to space travel. In order to escape the Earth's gravity from the surface, we require a velocity of 11.2 kilometers per second. To find out the amount of fuel required, there's a simple equation known as the 'Rocket Equation'. It tells us that an object can apply acceleration to itself





Although the experiment shows thrust, there's no clear explanation as to how the system works